



TAMGA NETWORK

A Digital Trust Infrastructure for Türkiye and the Turkic World

A sovereign reference architecture on the EU's EUDI profiles — X.509 institutions, signed trust lists today, a permissioned ledger when independent operators join.

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Building Trust Infrastructure for the Digital World.

Executive summary

Tamga Network is a **Digital Trust Infrastructure**: institutions issue documents — diplomas, student cards, identity credentials, tickets — into a person’s phone, the person shares only the fields a verifier needs, and the verifier checks them in seconds without contacting the issuer.

Tamga keeps the EU’s technical layer unchanged: **SD-JWT VC** and **ISO 18013-5 mdoc** credentials, **OpenID4VCI/VP**, **X.509** institutions and **signed trust lists** in the ETSI model. Its own layer is governance for the Turkic world: every structure has a slot for each member state, and Tamga acts only as a provisional operator on their behalf.

Trust is anchored today in versioned, hash-chained trust lists with a public anchor log; a permissioned **Besu/QBFT** ledger is added only when at least two independent operators join. No personal data — not even a hash — is written to lists, logs or ledger. The full flow runs end to end today with real cryptography; this document says what works, what is planned and what is still research.

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1 The problem: the end of the “document copy”

For decades people proved who they are by handing over copies. Copies pile up on servers and slip out of control; every bank, employer and university rebuilds the same verification; and authenticity is left to how convincing a copy looks.

The new model reverses this: the document stays with the person, only the necessary proof is shared, and verification is cryptographic — without asking the source. See [why a new model](#).

2 Vision: from eIDAS 2.0 to the Turkic world

With **eIDAS 2.0** every EU member state must offer its citizens a **European Digital Identity Wallet**. Its architecture (the ARF) and profiles are becoming the de-facto standard for digital trust.

The Turkic world shares language, culture and history. A diploma issued in one state should be verifiable in another; an institution’s identity should be trusted across borders. Tamga builds that shared foundation on the same standards, with governance that keeps every state sovereign.

3 Principles

- **Sovereignty first** — each state is the only writer of its own registry; network membership by a 2/3 vote; cross-border recognition decided unilaterally.
- **No personal data in any shared record** — not in lists, logs or ledger; not even a hash.
- **Compatible but independent** — the EU’s technical layer as it is; the governance layer written for the Turkic world.
- **Built for many states** — no identifier or role assumes Tamga as the only operator; Tamga is always the provisional stand-in.
- **A ledger is a choice of signers, not of storage** — lists today; a ledger when independent operators join.
- **Holder binding without exception** — every credential copy is bound to a key on the device; a copy cannot be replayed.
- **Designed to be handed over** — every provisional power has a measurable handover point.

4 Roles

Every role of the EU architecture exists in Tamga. Where a state has not joined, Tamga holds the role provisionally and on the record: trusted-list operator, registrar, the “TR National Root CA (provisional operator: Tamga)” and the wallet provider. Institutions are attestation providers; employers, websites and gates are registered relying parties. The PID-provider slot is empty until a state fills it; meanwhile an identity credential comes from Tamga’s identity service (document and liveness check). There are no validator operators yet — hence no ledger. Side-by-side: [Tamga and the EUDI architecture](#).

5 Trust model: signed trust lists

A signature proves who signed; a **trust list** says whether that signer is a real institution, which document types it may issue, since when, and its current status. Lists are signed JWS files, **versioned and hash-chained**, never deleted, and carry a next-update date; verifiers check the signer against a root fingerprint published out of band. Every revocation-list publication and schema change is also written, at least hourly, to a public **anchor log**, so a rolled-back list can be detected. Details: [trust lists](#).

PUBLISHED FILES — TRUST.TAMGA.NETWORK

lotl.jws	list of lists — national lists, schemas, wallet providers
tl-tr.jws	Türkiye — root CAs, issuers (+ authorizations), relying parties
tl-az · kz · kg · uz	reserved slots for the other member states
anchors.jsonl	anchor log — one signed line per event, at least hourly
keys/	root fingerprints (the out-of-band trust anchor)
archive/	every past version, never deleted

Identifiers are derived, not assigned, and do not change on handover or when the ledger arrives:

IDENTIFIERS

ca_id	keccak256(state_code SHA-256(root certificate))
issuer_id	keccak256(state_code SHA-256(issuer certificate))
vct	urn:tamga:<domain>:<Type>:<major> — e.g. urn:tamga:edu:DiplomaCredential:1
schema_id	keccak256(vct)
person	no identifier — a device key per credential copy

6 Credentials: SD-JWT VC and mdoc

The main format is **SD-JWT VC** (IETF, dc+sd-jwt, ES256). Each field is hidden behind a salted hash and revealed only with the holder's approval; the header carries the institution's X.509 chain; cnf binds the copy to a device key. The identity credential is also issued as an **ISO 18013-5 mdoc**, so an age check can receive **age_over_18** and nothing else. Document types are stable URNs; their definitions sit in a public catalogue and every credential carries a hash of its definition. See [credentials](#).

AN SD-JWT VC DIPLOMA (DECODED, SHORTENED)

```
{
  "iss": "https://issuer.tamga.network/example-university",
  "vct": "urn:tamga:edu:DiplomaCredential:1",
  "vct#integrity": "sha256-...",
  "iat": 1790000000,
  "cnf": { "jwk": { ... } },
  "status": { "status_list": {
    "idx": 48213,
    "uri": "https://status.tamga.network/..." } },
  "_sd": [ "...", "..." ]
}
```

cnf.jwk	device key of this copy
_sd	hidden fields, as salted hashes
header · x5c	the institution's X.509 certificate chain

The national ID number appears only in the identity credential; no diploma, card or ticket carries it.

7 Issuing and presenting

- **Issuance (OpenID4VCI).** The institution shows a QR code with a PIN on the same screen — the PIN never travels inside the link. Or the wallet starts from the institution directory and proves the person’s identity first. The wallet receives **ten copies**, each bound to a different device key.
- **Genuine wallets only.** Issuers require a short-lived **wallet unit attestation** from the wallet provider.
- **Presentation (OpenID4VP, DCQL).** The verifier’s request is signed with its registered certificate. The wallet checks it against the trust list, shows exactly what is asked and warns about anything beyond the verifier’s registered scope, then sends an encrypted answer with the approved fields and a proof that the key is on this phone.
- **Websites.** A site signs a person up once with the wallet; daily sign-in then uses a **passkey** and shares no fields. See [sign in with TamgaID](#).

8 Verification: five layers, three outcomes

Every verification runs the same pipeline in the same order and stops at the first failure. Each step has a permanent code, so a rejection always states why.

VERIFICATION PIPELINE

T0	request and answer belong together (nonce, audience, encryption)
A · format	signature, certificate chain, device proof, hidden fields intact
B · type	document type registered; definition hash matches the catalogue
C · trust	issuer authorized for this type on the issue date; category matches
D · status	not revoked or suspended; list fresh and anchored
E · policy	requested fields present; nothing beyond the verifier's scope
→ outcome	ACCEPTED · REJECTED (failing step) · INDETERMINATE (could not check)

INDETERMINATE is never reported as **REJECTED**. If a list cannot be reached or is out of date, the verifier says “could not check right now” — the difference between “this diploma is fake” and “I cannot check” decides whether someone is hired. Authorization is judged on the **issue date**: a diploma issued while a university was active stays valid after a suspension, while new issuance stops at once.

9 Revocation and lifecycle

Revocation uses the **IETF Token Status List**: two bits per credential copy — valid, revoked or suspended — at a **random** position. The issuer publishes at a **fixed interval**, never on demand, so timing reveals nothing about a person; each publication is anchored. Verifiers pre-fetch the lists, so checking a credential makes no call to the issuer or to the phone. A revocation reaches every verifier within about 90 minutes at most. Copies run out by design; the wallet asks before fetching fresh ones and never refreshes silently. See [recovery and revocation](#).

10 Privacy by design

- **Per-verifier copies.** Each verifier receives a different copy bound to a different key, so verifiers cannot link a person by comparing what they received.
- **Selective disclosure** by default; predicates such as `age_over_18` where the format allows.
- **Identity checks are isolated.** Only the identity service talks to the identity-verification provider; after issuing it keeps no photos, only an opaque hashed record (a subject reference and a document-number hash). Institutions match a person through the identity credential, not through the provider.
- **No personal data in logs or public addresses.** Revocation-list addresses never encode the institution; logs record what happened, never to whom.
- **Websites** store their own keyed hash of the account key, never the raw value.

Residual risk, stated openly: the same issuer colluding with several verifiers could still link a person. Closing this requires zero-knowledge credentials (see research directions).

11 Close range: passes and age checks

For turnstiles and event gates Tamga uses a **pass**: registration once through a standard presentation, then a 60-second signed token shown as a QR code — no personal data in the token, replay rejected, and tickets can be single-use. A person-to-person check reverses OpenID4VP so the checker's app starts a standard request. Both are versioned bridges; the target is ISO 18013-5 over NFC/BLE.

12 From lists to a ledger

A ledger adds something only when several independent parties run it. Tamga therefore starts with lists and adds a permissioned **Hyperledger Besu** network with **QBFT** consensus only once at least two independent validator operators agree in writing. Every list field maps to a contract record; the list history is replayed into the contracts and both are tested to give the same answers. Components read trust through one interface, so documents, wallets and the verification pipeline do not change. See [what blockchain is — and isn't](#).

Governance on the ledger follows the principles: validators are states with equal votes; new members by a 2/3 vote; each state alone registers or suspends its own institutions; recognition of foreign institutions is decided by each state.

13 Research directions

These are designs under study, not part of the first release or the pilot; each will pass an independent security review before any use.

- **Zero-knowledge credentials** (e.g. BBS+) to remove issuer linkability. See [selective disclosure](#).
- **Accountable disclosure** — pseudonyms resolvable only by a court plus a multi-institution threshold (DKG and threshold ElGamal; the key is never reconstructed). See [accountable disclosure](#).
- **Derived pseudonyms and recovery** — hierarchical keys, envelope encryption, device recovery. See [identity layers](#).
- **The value layer** — authorization between verified parties; settlement stays on regulated rails.

14 Status and roadmap

Working today (first release, real cryptography): issuance and presentation of diplomas and student cards; revocation and institution suspension; identity check and identity credential, also as mdoc; campus and event passes, single-use tickets; website sign-up and passkey sign-in; eight open-source packages. Tested on a phone.

PHASES

Phase B (today)	signed trust lists + anchor log · Tamga = provisional operator
Pilot	one foundation university · issuer key at the university · lists, no ledger
Phase 0	permissioned Besu/QBFT ledger once at least 2 independent validator operators sign
Phase 1	member-state lists · close range (NFC/BLE) · Digital Credentials API

Every first-release shortcut — sample records, software keys, the issuer key held by Tamga, a single operator — is listed in a public deviation log and closed before the pilot. The pilot’s success and stop criteria are defined in advance.

15 Known limits

- In this phase the trust anchor rests on one operator’s signature. The public log, transparency report and audits deter misuse; they cannot make it impossible.
- A revocation takes effect within about 90 minutes at most.
- Issuer linkability remains until zero-knowledge credentials are adopted.
- First release only: keys in software and the issuer key held by Tamga — both close before the pilot.
- Until Tamga Wallet is on the App Store and Google Play, the phone’s own claim of secure hardware is not accepted; with the store release App Attest / Play Integrity become mandatory.

16 Open source and further reading

Code is Apache-2.0 and documentation CC BY 4.0. The packages `@tamga-network/*` cover trust lists, credential formats, issuing, verification and the wallet core; the integration guides are in the [developer docs](#). Concepts are explained from scratch in the [documentation](#) the [glossary](#) is a quick reference and the [manifesto](#) gives the “why”.

This is a living document (v3.0). It changes as decisions mature; the trust model is what stays.